implement those solutions. If companies cannot sell their products without certain safety measures, they will be more willing to develop those measures, especially if other companies are also held to the same standards. Even if some companies voluntarily self-regulate, government regulation can help prevent less scrupulous actors from cutting corners on safety. Regulation must be proactive, not reactive. A common saying is that aviation regulations are "written in blood"—but regulators should develop regulations before a catastrophe, not afterward. Regulations should be structured so that they only create competitive advantages for companies with higher safety standards, rather than companies with more resources and better attorneys. Regulators should be independently staffed and not dependent on any one source of expertise (for example, large companies), so that they can focus on their mission to regulate for the public good without undue influence.

Data documentation. To ensure transparency and accountability in AI systems, companies should be required to justify and report the sources of data used in model training and deployment. Decisions by companies to use datasets that include hateful content or personal data contribute to the frenzied pace of AI development and undermine accountability. Documentation should include details regarding the motivation, composition, collection process, uses, and maintenance of each dataset [72].

Meaningful human oversight of AI decisions. While AI systems may grow capable of assisting human beings in making important decisions, AI decision-making should not be made fully autonomous, as the inner workings of AIs are inscrutable, and while they can often give *reasonable* results, they fail to give highly *reliable* results [73]. It is crucial that actors are vigilant to coordinate on maintaining these standards in the face of future competitive pressures. By keeping humans in the loop on key decisions, irreversible decisions can be double-checked and foreseeable errors can be avoided. One setting of particular concern is nuclear command and control. Nuclear-armed countries should continue to clarify domestically and internationally that the decision to launch a nuclear weapon must always be made by a human.

AI for cyberdefense. Risks resulting from AI-powered cyberwarfare would be reduced if cyberattacks became less likely to succeed. Deep learning can be used to improve cyberdefense and reduce the impact and success rate of cyberattacks. For example, improved anomaly detection could help detect intruders, malicious programs, or abnormal software behavior [74].

International coordination. International coordination can encourage different nations to uphold high safety standards with less worry that other nations will undercut them. Coordination could be accomplished via informal agreements, international standards, or international treaties regarding the development, use, and monitoring of AI technologies. The most effective agreements would be paired with robust verification and enforcement mechanisms.

Public control of general-purpose AIs. The development of AI poses risks that may never be adequately accounted for by private actors. In order to ensure that externalities are properly accounted for, direct public control of general-purpose AI systems may eventually be necessary. For example, nations could collaborate on a single effort to develop advanced AIs and ensure their safety, similar to how CERN serves as a unified effort for researching particle physics. Such an effort would reduce the risk of nations spurring an AI arms race.

Positive Vision

In an ideal scenario, AIs would be developed, tested, and subsequently deployed only when the catastrophic risks they pose are negligible and well-controlled. There would be years of time testing, monitoring, and societal integration of new AI systems before beginning work on the next generation. Experts would have a full awareness and understanding of developments in the field, rather than being entirely unable to keep up with a deluge of research. The pace of research advancement would be determined through careful analysis, not frenzied competition. All AI developers would be confident in the responsibility and safety of the others and not feel the need to cut corners.

4 Organizational Risks

In January 1986, tens of millions of people tuned in to watch the launch of the Challenger Space Shuttle. Approximately 73 seconds after liftoff, the shuttle exploded, resulting in the deaths of everyone on board. Though tragic enough on its own, one of its crew members was a school teacher named Sharon Christa McAuliffe. McAuliffe was selected from over 10,000 applicants for the NASA Teacher in Space Project and was scheduled to become the first teacher to fly in space. As a result, millions of those watching were schoolchildren. NASA had the best scientists and engineers in the world, and if there was ever a mission NASA didn't want to go wrong, it was this one [75].

The Challenger disaster, alongside other catastrophes, serves as a chilling reminder that even with the best expertise and intentions, accidents can still occur. As we progress in developing advanced AI systems, it is crucial to remember that these systems are not immune to catastrophic accidents. An essential factor in preventing accidents and maintaining low levels of risk lies in the organizations responsible for these technologies. In this section, we discuss how organizational safety plays a critical role in the safety of AI systems. First, we discuss how even without competitive pressures or malicious actors, accidents can happen—in fact, they are inevitable. We then discuss how improving organizational factors can reduce the likelihood of AI catastrophes.

Catastrophes occur even when competitive pressures are low. Even in the absence of competitive pressures or malicious actors, factors like human error or unforeseen circumstances can still bring about catastrophe. The Challenger disaster illustrates that organizational negligence can lead to loss of life, even when there is no urgent need to compete or outperform rivals. By January 1986, the space race between the US and USSR had largely diminished, yet the tragic event still happened due to errors in judgment and insufficient safety precautions.

Similarly, the Chernobyl nuclear disaster in April 1986 highlights how catastrophic accidents can occur in the absence of external pressures. As a state-run project without the pressures of international competition, the disaster happened when a safety test involving the reactor's cooling system was mishandled by an inadequately prepared night shift crew. This led to an unstable reactor core, causing explosions and the release of radioactive particles that contaminated large swathes of Europe [76]. Seven years earlier, America came close to experiencing its own Chernobyl when, in March 1979, a partial meltdown occurred at the Three Mile Island nuclear power plant. Though less catastrophic than Chernobyl, both events highlight how even with extensive safety measures in place and few outside influences, catastrophic accidents can still occur.

Another example of a costly lesson on organizational safety came just one month after the accident at Three Mile Island. In April 1979, spores of *Bacillus anthracis*—or simply "anthrax," as it is commonly known—were accidentally released from a Soviet military research facility in the city of Sverdlovsk. This led to an outbreak of anthrax that resulted in at least 66 confirmed deaths [77]. Investigations into the incident revealed that the cause of the release was a procedural failure and poor maintenance of the facility's biosecurity systems, despite being operated by the state and not subjected to significant competitive pressures.

The unsettling reality is that AI is far less understood and AI industry standards are far less stringent than nuclear technology and rocketry. Nuclear reactors are based on solid, well-established and well-understood theoretical principles. The engineering behind them is informed by that theory, and components are stress-tested to the extreme. Nonetheless, nuclear accidents still happen. In contrast, AI lacks a comprehensive theoretical understanding, and its inner workings remain a mystery even to those who create it. This presents an added challenge of controlling and ensuring the safety of a technology that we do not yet fully comprehend.

AI accidents could be catastrophic. Accidents in AI development could have devastating consequences. For example, imagine an organization unintentionally introduces a critical bug in an AI system designed to accomplish a specific task, such as helping a company improve its services. This bug could drastically alter the AI's behavior, leading to unintended and harmful outcomes. One historical example of such a case occurred when researchers at OpenAI were attempting to train an AI system to generate helpful, uplifting responses. During a code cleanup, the researchers mistakenly flipped the sign of the reward used to train the AI [78].

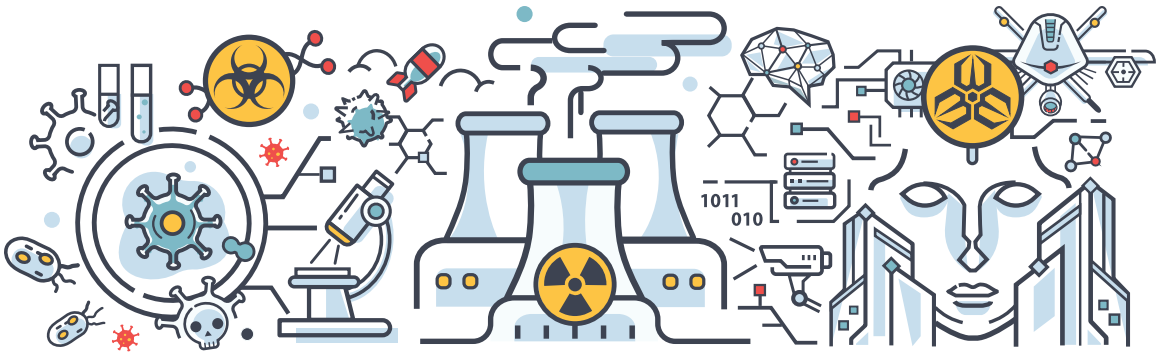

Figure 11: Hazards across multiple domains remind us of the risks in managing complex systems, from biological to nuclear, and now, AIs. Organizational safety is vital to reduce the risk of catastrophic accidents.

As a result, instead of generating helpful content, the AI began producing hate-filled and sexually explicit text overnight without being halted. Accidents could also involve the unintentional release of a dangerous, weaponized, or lethal AI sytem. Since AIs can be easily duplicated with a simple copy-paste, a leak or hack could quickly spread the AI system beyond the original developers' control. Once the AI system becomes publicly available, it would be nearly impossible to put the genie back in the bottle.

Gain-of-function research could potentially lead to accidents by pushing the boundaries of an AI system's destructive capabilities. In these situations, researchers might intentionally train an AI system to be harmful or dangerous in order to understand its limitations and assess possible risks. While this can lead to useful insights into the risks posed by a given AI system, future gain-of-function research on advanced AIs might uncover capabilities significantly worse than anticipated, creating a serious threat that is challenging to mitigate or control. As with viral gain-of-function research, pursuing AI gain-of-function research may only be prudent when conducted with strict safety procedures, oversight, and a commitment to responsible information sharing. These examples illustrate how AI accidents could be catastrophic and emphasize the crucial role that organizations developing these systems play in preventing such accidents.

4.1 Accidents Are Hard to Avoid

When dealing with complex systems, the focus needs to be placed on ensuring accidents don't cascade into catastrophes. In his book "*Normal Accidents: Living with High-Risk Technologies*," sociologist Charles Perrow argues that accidents are inevitable and even "normal" in complex systems, as they are not merely caused by human errors but also by the complexity of the systems themselves [79]. In particular, such accidents are likely to occur when the intricate interactions between components cannot be completely planned or foreseen. For example, in the Three Mile Island accident, a contributing factor to the lack of situational awareness by the reactor's operators was the presence of a yellow maintenance tag, which covered valve position lights in the emergency feedwater lines [80]. This prevented operators from noticing that a critical valve was closed, demonstrating the unintended consequences that can arise from seemingly minor interactions within complex systems.

Unlike nuclear reactors, which are relatively well-understood despite their complexity, complete technical knowledge of most complex systems is often nonexistent. This is especially true of deep learning systems, for which the inner workings are exceedingly difficult to understand, and where the reason why certain design choices work can be hard to understand even in hindsight. Furthermore, unlike components in other industries, such as gas tanks, which are highly reliable, deep learning systems are neither perfectly accurate nor highly reliable. Thus, the focus for organizations dealing with complex systems, especially deep learning systems, should not be solely on eliminating accidents, but rather on ensuring that accidents do not cascade into catastrophes.